

Work Flow Diagrams

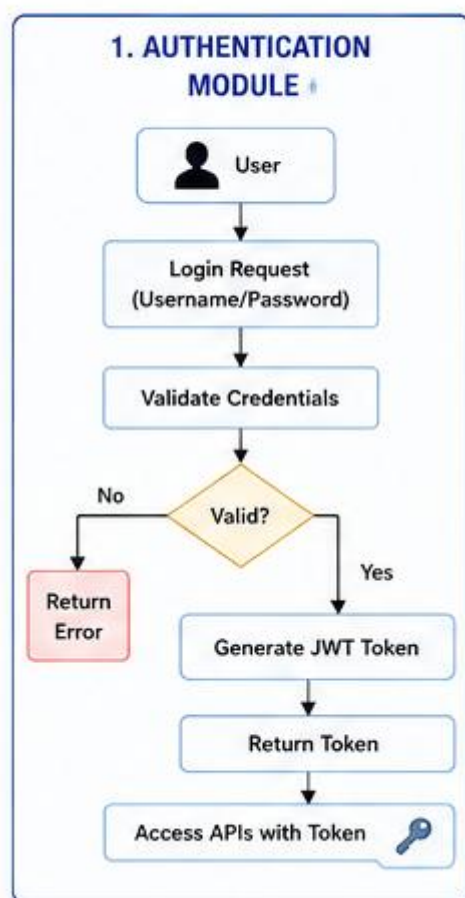
5.1 Authentication Module

Description:

The Authentication Module verifies user credentials and provides secure access to the system. After successful login, a JWT token is generated, allowing users to access protected resources securely.

Workflow

- [?] User enters login credentials.
- [?] System validates the user.
- [?] JWT token is generated.
- [?] User accesses secured modules.Invalid credentials



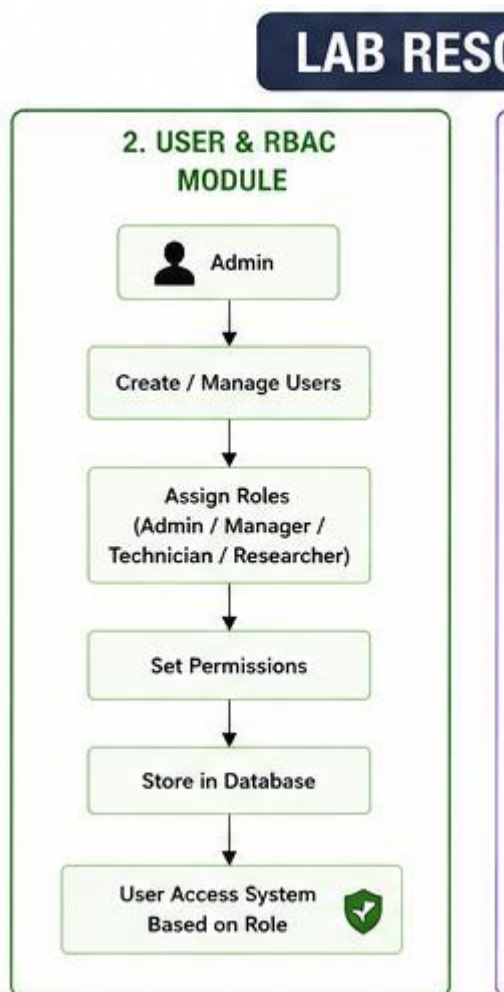
5.2 User & RBAC Module

Description:

This module manages users, roles, and permissions. Role-Based Access Control (RBAC) ensures users can only access features permitted by their assigned role.

Workflow

- Create user.
- Assign role.
- Configure permissions.
- Grant authorized access.



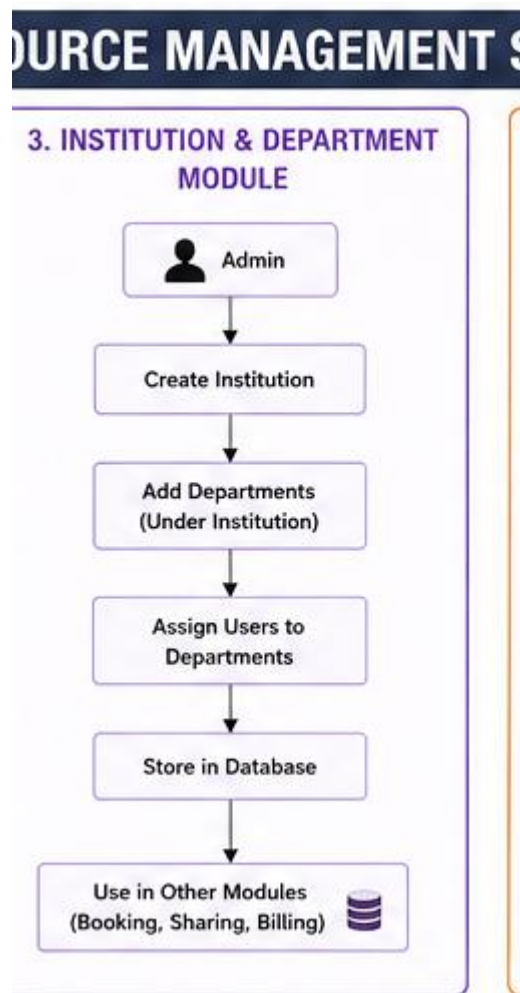
5.3 Institution&Department Management Module

Description

This module manages institution information and maintains records of participating organizations. It serves as the foundation for department and laboratory management.

Workflow

- Create institution.
- Update details.
- View information.
- Store in database.



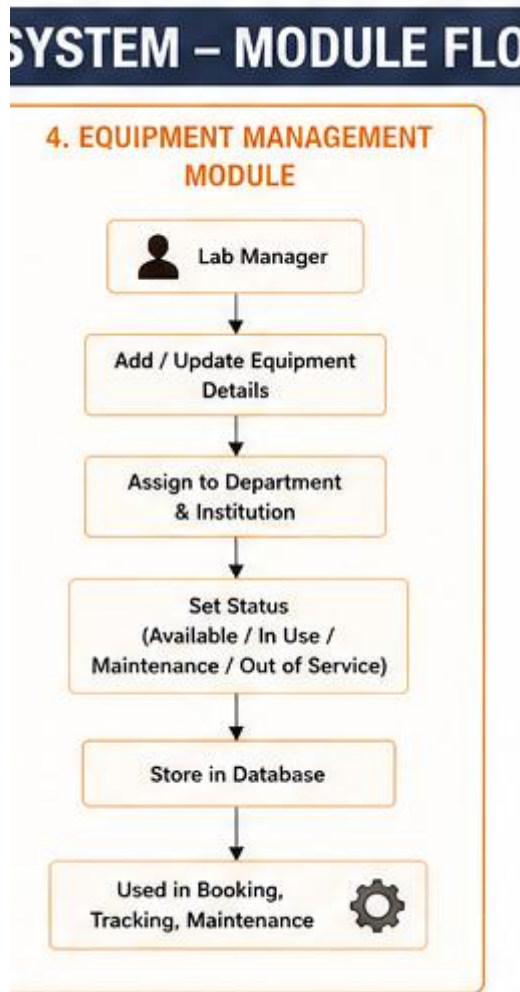
5.5 Equipment Management Module

Description

This module manages laboratory equipment, including its details, availability, and current status for efficient resource utilization.

Workflow

- Add equipment.
- Update equipment.
- Check availability.
- Store information.



5.6 Booking Management Module

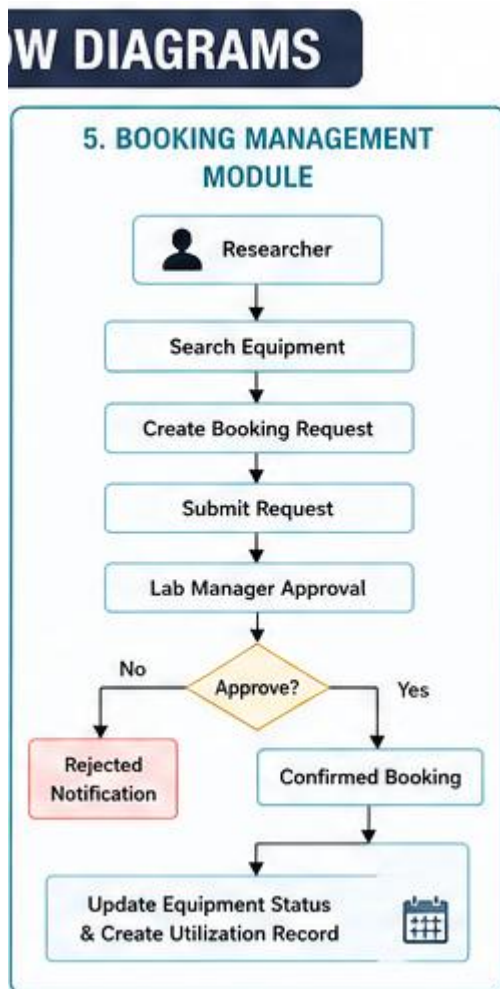
Description

The Booking Module allows users to reserve laboratory equipment after verifying its availability and obtaining approval when required.

Workflow

- Search equipment.

- Create booking.
- Check availability.
- Confirm booking.



5.7 Resource Sharing Module

Description

This module enables institutions to share laboratory equipment with other institutions through an approval process.

Workflow

- Create sharing request.
- Validate request.
- Approve or reject.

- Update sharing status.

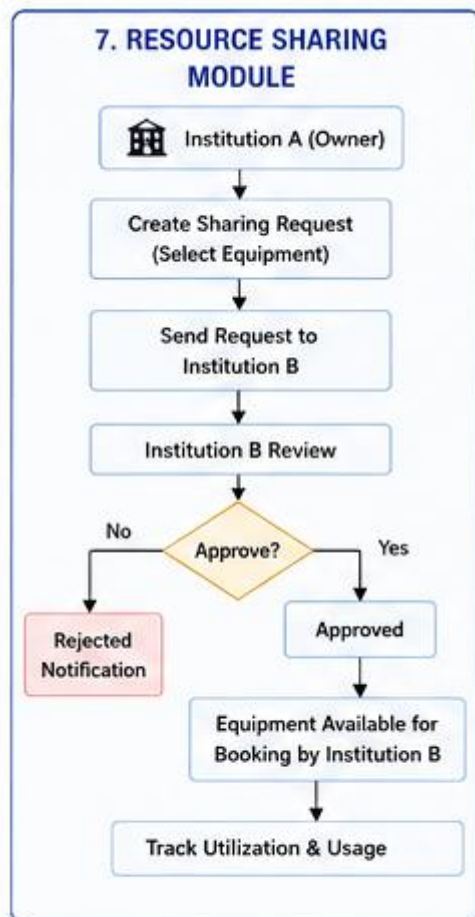
5.8 External Booking Module

Description

The External Booking Module supports equipment reservations by external institutions after resource-sharing approval.

Workflow

- Request booking.
- Verify sharing.
- Create booking.
- Confirm reservation.



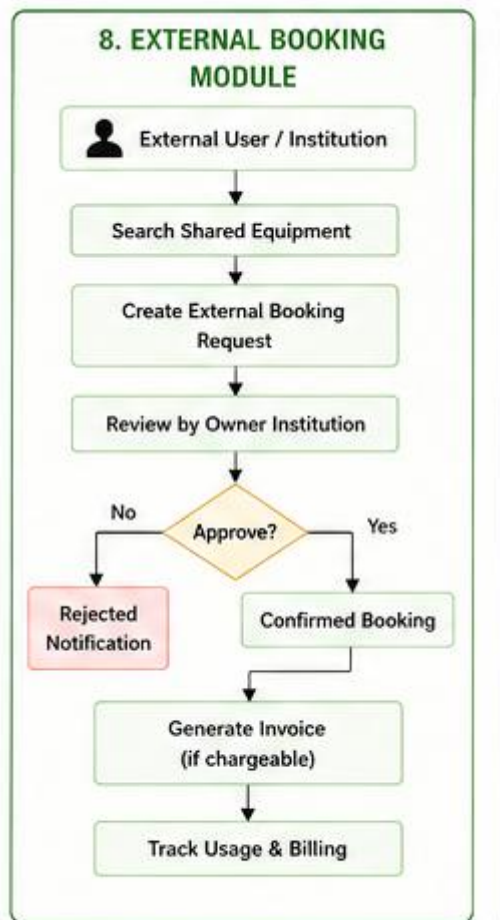
5.8 External Booking Module

Description

The External Booking Module supports equipment reservations by external institutions after resource-sharing approval.

Workflow

- Request booking.
- Verify sharing.
- Create booking.
- Confirm reservation.



5.9 Maintenance Management Module

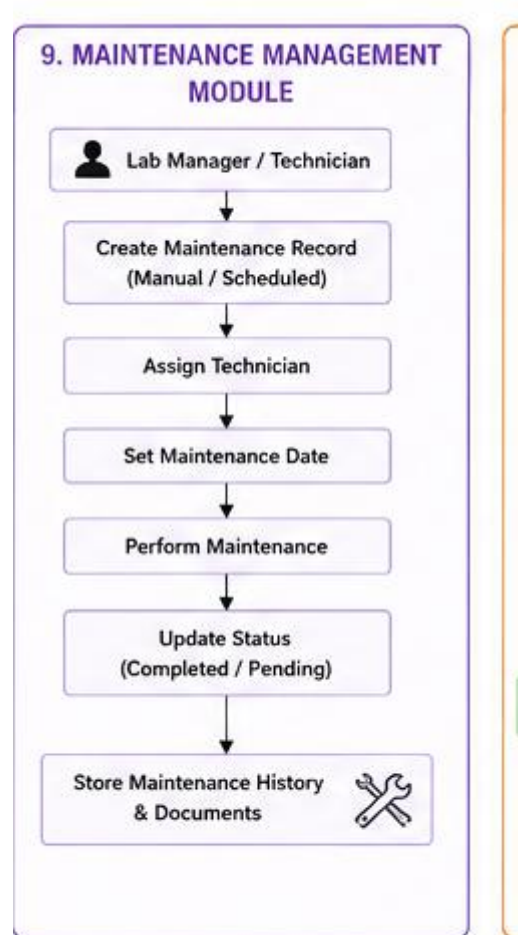
Description

This module records and manages equipment maintenance activities to ensure proper functioning and improve equipment life.

Workflow

- Schedule maintenance.

- Assign technician.
- Update status.
- Save maintenance record.



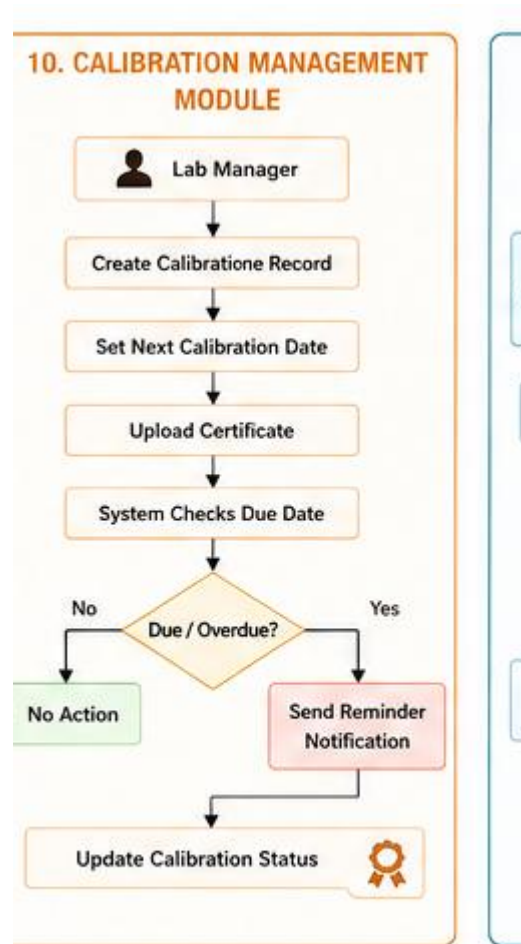
5.10 Calibration Management Module

Description

The Calibration Module maintains calibration schedules and records to ensure equipment accuracy and reliability.

Workflow

- Schedule calibration.
- Perform calibration.
- Save record.
- Update status.



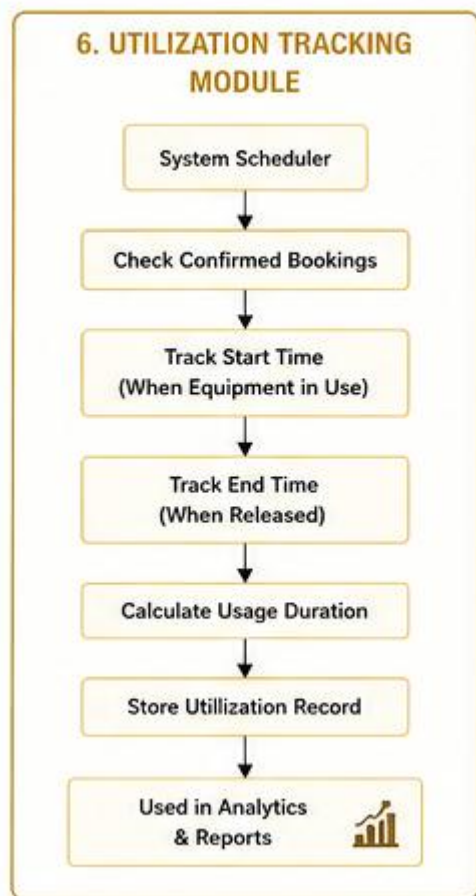
5.11 Utilization Tracking Module

Description

This module tracks equipment usage and calculates utilization statistics for better resource planning and analysis.

Workflow

- Record usage.
- Calculate duration.
- Update utilization.
- Generate statistics.



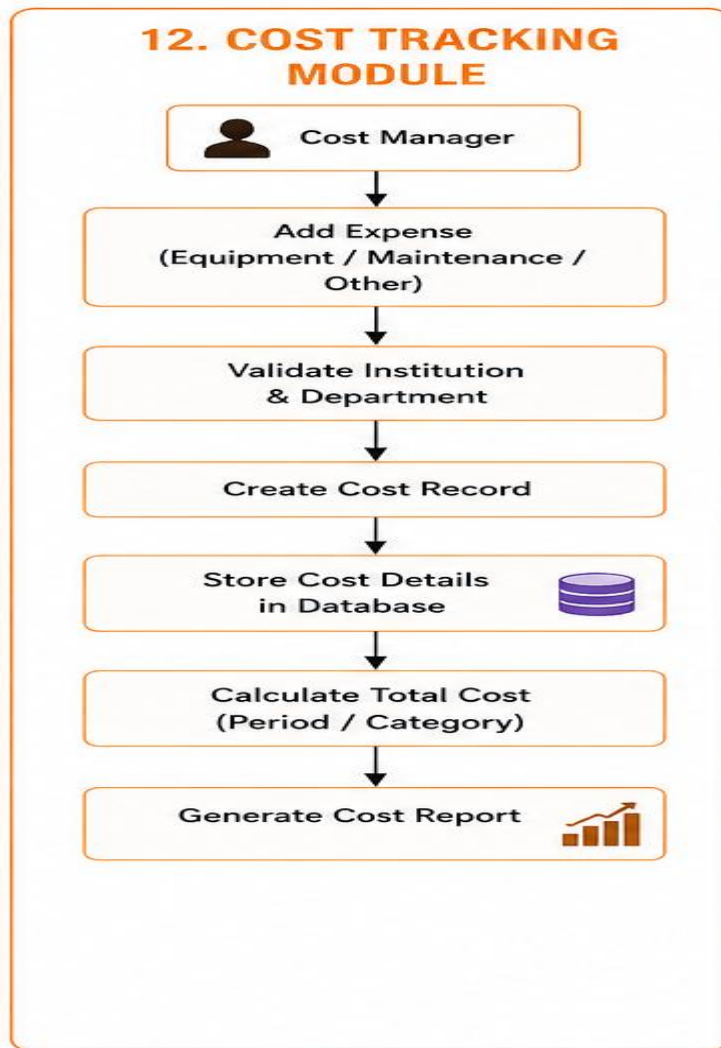
5.12 Cost Tracking Module

Description

The Cost Tracking Module records expenses related to equipment, maintenance, and laboratory operations for financial monitoring.

Workflow

- Add expense.
- Calculate cost.
- Save record.



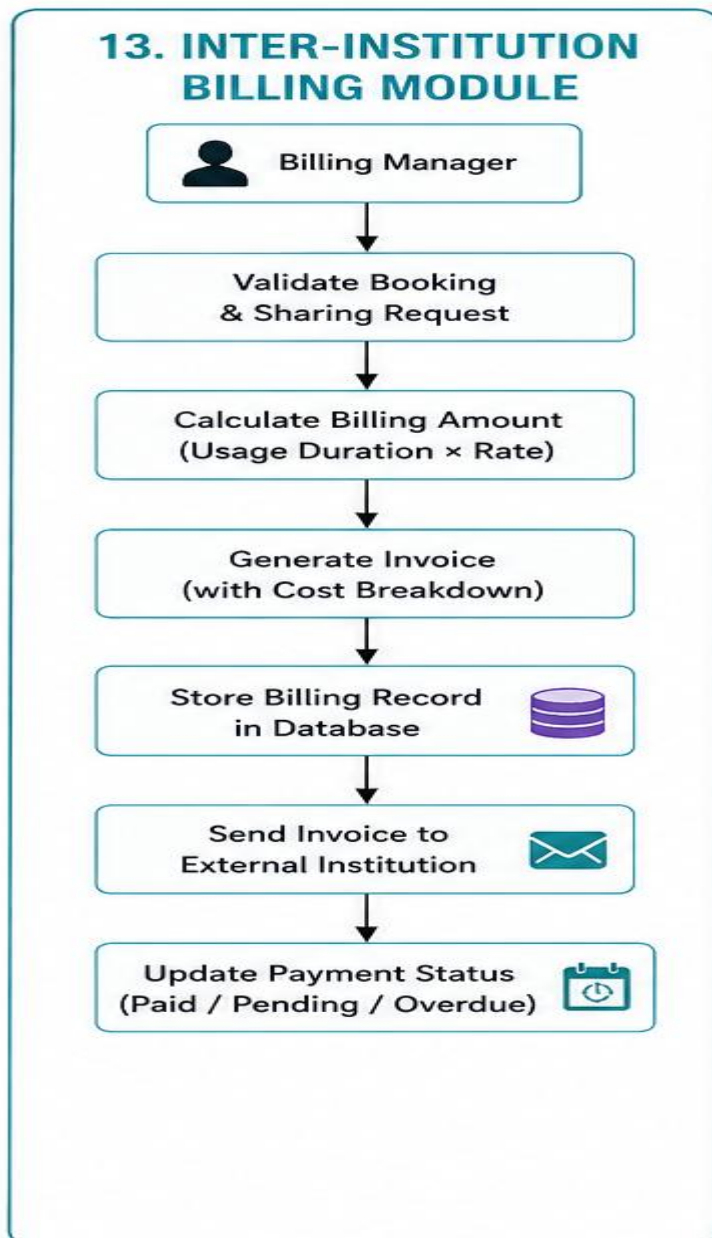
5.13 Inter-Institution Billing Module

Description

This module generates invoices for shared resources and tracks payments between collaborating institutions.

Workflow

- Generate invoice.
- Calculate charges.
- Track payment.
- Update billing status.



5.14 Reports Module

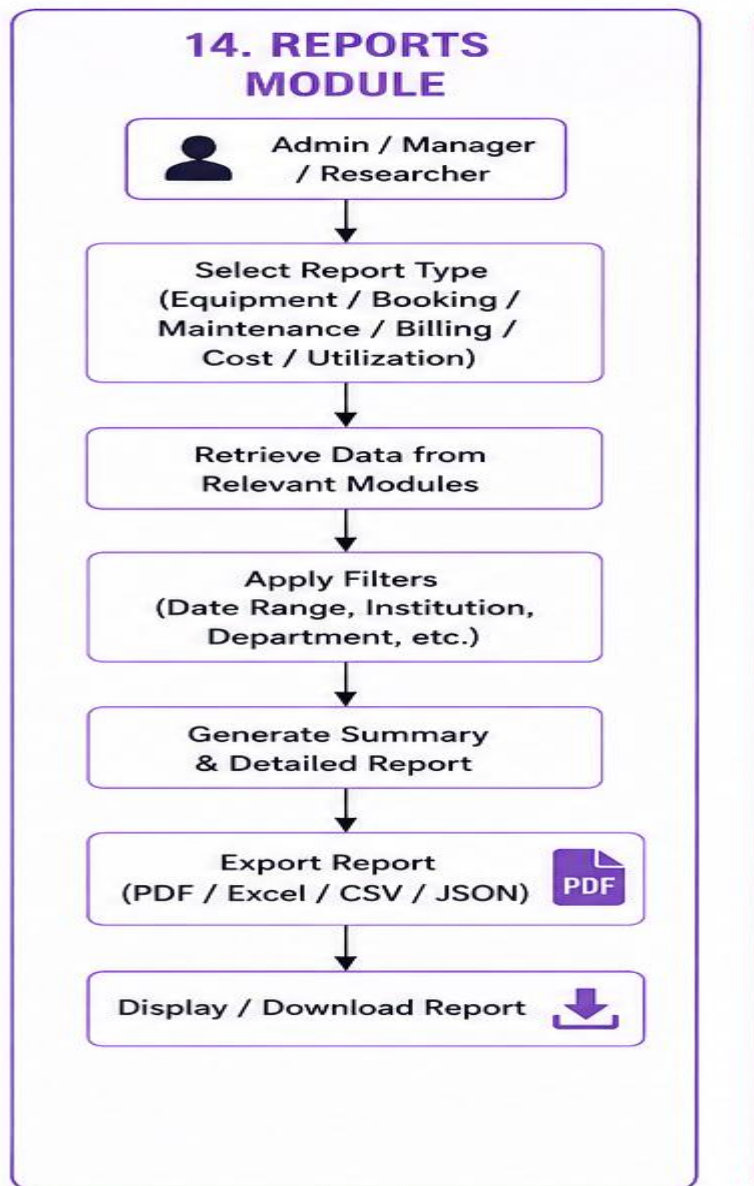
Description

The Reports Module generates summaries and detailed reports for equipment, bookings, maintenance, billing, and utilization.

Workflow

- Collect data.
- Generate report.
- Export report.

- Display results.



5.15 Dashboard & Analytics Module

Description

The Dashboard displays key statistics and analytics, providing administrators with a quick overview of laboratory operations.

Workflow

- Collect data.
- Analyze information.

- Generate statistics.
- Display dashboard.

